FACULTY
OF INFORMATION
TECHNOLOGY
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# Assignment of bachelor's thesis

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| **Title:** | Self-Organizing Maps and Their Modern Extensions |
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| **Supervisor:** | Ing. Magda Friedjungová, Ph.D. |
| **Study program:** | Informatics |
| **Branch / specialization:** | Artificial Intelligence 2021 |
| **Department:** | Department of Applied Mathematics |
| **Validity:** | until the end of summer semester 2026/2027 |

## Instructions

A self-organizing map (SOM) is an unsupervised machine learning method used to produce a low-dimensional (typically two-dimensional) representation of a higher-dimensional dataset while preserving the topological structure of the data. The aim of this thesis is extensively explore current state of SOM-based methods and provide various visualizations of their results.

1. Survey classical and state-of-the-art SOM-based methods, with emphasis on representation learning, visualization techniques, and application domains.
2. Implement or adapt at least three surveyed methods and apply them to publicly available datasets.
3. Evaluate the selected methods using appropriate quantitative measures and visualization techniques, and compare their ability to preserve topological relationships.

*Electronically approved by Ing. Karel Klouda, Ph.D. on 14 January 2026 in Prague.*